

Measuring & Mitigating AI Over-Reliance in Programming Education

A Telemetry-Driven Behavioral Metric with Productive Friction on S-SPARC & E-STRANGE

RESEARCH TEAM

S-SPARC Project Team

TARGET FIELD

AIED & Learning Analytics

PLATFORM

S-SPARC & E-STRANGE
LMS

PUBLICATION TARGET

IEEE TALE / Scopus Q1

EXECUTIVE SUMMARY

Generative AI can generate working code in seconds, but unchecked student access often leads to cognitive offloading and surface-level learning. This research project introduces the **AI Over-Reliance Index (ORI)**: an automated, real-time behavioral metric built into the S-SPARC AI engine and E-STRANGE auto-grader. By logging unobtrusive interaction telemetry and enforcing structured productive friction (60-second reflection pauses), this study provides higher education instructors with an empirical framework to diagnose and prevent cognitive atrophy without banning AI tools.

1. The Problem: Cognitive Offloading in Programming Labs

In programming classes, students frequently fall into a passive interaction loop: copying problem prompts into AI, copying generated code, and submitting it directly to automated graders. While lab assignment scores look high, students struggle during unassisted exams when AI is not available [2], [3].

- **Critical Thinking Decay:** Literature shows excessive reliance on conversational AI degrades independent problem-solving and self-debugging skills [2].
- **Weak Conceptual Transfer:** Instant AI assistance shows minimal correlation with genuine conceptual mastery [3], [14].
- **Impact on Course Outcomes:** High LLM dependency is negatively correlated with final exam scores [4].
- **Superficial Engagement:** Studies of student-ChatGPT dialogues show students rarely reflect deeply on the code they receive [5].

2. Why Existing Solutions Fall Short

Current Method	Main Limitation	S-SPARC Solution
Surveys / Questionnaires [2], [7]	Subjective, prone to recall bias, cannot measure live behavior.	Continuous Telemetry: Automatic logging of typing speed, pauses, and prompt quality per millisecond.
AI Plagiarism Checkers	High false-positive rate, purely punitive, zero learning feedback.	Cognitive Profiling: Maps student problem-solving habits over time.
Lab Eye-Tracking [16]	Rigorous but post-hoc and only works in specialized lab setups.	Real-Time Metric (ORI): Computes live reliance scores to adjust AI scaffolding automatically.
Unregulated AI Tools [5]	Commercial chatbots encourage rapid prompt spamming without thought.	Productive Friction: Mandatory 60s pause and stepped Socratic hints [9], [10].

3. The Proposed Metric: AI Over-Reliance Index (ORI)

The ORI combines four unobtrusive telemetry indicators into a normalized composite score from 0.0 (independent problem solver) to 1.0 (heavily over-reliant):

$$\text{ORI} = 1 - (w1 \cdot Q_{\text{prompt}} + w2 \cdot S_{\text{CIOE}} + w3 \cdot R_{\text{concept}} + w4 \cdot F_{\text{reflect}})$$

All components normalized between 0 and 1 • Baseline weights: w1=0.30, w2=0.25, w3=0.20, w4=0.25

1. Shannon Prompt Entropy (Q_prompt)

Uses Shannon Information Entropy $H(X) = -\sum p(x) \log_2 p(x)$ [11] to measure technical vocabulary richness and specificity in student queries [13], [17]. Penalizes vague prompts like "why error".

2. Context Completeness (S_CIOE)

Checks compliance across four structured dimensions: Context, Input specs, Expected Output, and Error Trace [12]. Rewards well-formulated problem descriptions.

3. Conceptual Ratio (R_concept)

Measures the proportion of queries asking for explanations/logic (Bloom C2-C4) versus requests for instant code generation (Bloom C1).

4. Reflection Adherence (F_reflect)

Tracks adherence to the 60-second reflection pause [9] combined with keystroke cadence analysis to detect and penalize instant copy-pasting from external AI tools.

4. Methodological Defenses & Research Rigor

To ensure the study meets top-tier peer-review standards (IEEE / Scopus Q1), the research design explicitly addresses key threats to validity:

- **Reactivity & Maturation Confound:** The study uses an A-B design (Modules 1-2 baseline without friction, Modules 3-4 with friction). If only 1 cohort is available, the maturation effect of time is

transparently acknowledged as a limitation in the paper; if 2 sections exist, counterbalancing (A-B vs B-A) is applied.

- **Two-Phase Psychometric Roadmap:** To prevent sample size issues and factor overfitting, the research is split into two phases:
 - **Phase 1 (Semester 1 / Cohort 1, N=35-50):** Instrument development and Exploratory Factor Analysis (EFA) to discover factor structure and calibrate initial weights.
 - **Phase 2 (Semester 2 / Cohort 2, N ≥ 80):** Independent Confirmatory Factor Analysis (CFA), construct validation against the standardized Hou et al. (2025 [1]) scale, and testing interaction models.
- **Entropy Inversion Paradox:** Advanced students write brief, concise prompts because they know exactly what they need. We model an interaction term (`Entropy × Prior Ability`) so expert conciseness is not misclassified as over-reliance.
- **NLP Classifier Validation:** Intent classification for `R_concept` is validated on ≥ 100 sample prompts with double annotation by 2 independent raters, requiring Cohen's Kappa $\kappa \geq 0.80$.
- **Ethical Masking (Anti-Pygmalion Bias):** Individual student ORI scores and personas are blind-masked from course lecturers during Phase 1 to prevent subjective grading bias. Live lecturer dashboards unlock in Phase 2 post-validation.
- **Anti P-Hacking & Anti-Circularity:** The analysis plan is pre-registered on OSF.io. Bobot regresi diuji menggunakan 5-Fold Cross-Validation, dan 4 hipotesis eksploratori dikoreksi dengan Bonferroni ($\alpha = 0.0125$).

5. Implementation Timeline (8 Weeks to Submission)

Timeline	Activities	Deliverables
Weeks 1 - 2	OSF Pre-Registration, Ethics Approval, Baseline Phase (Modules 1-2)	Locked OSF protocol, baseline interaction logs
Weeks 3 - 4	Intervention Phase (Modules 3-4 with active 60s friction)	Complete telemetry dataset
Weeks 5 - 6	MySQL ETL extraction, NLP double annotation ($\kappa \geq 0.80$), EFA analysis	Psychometric factor report
Weeks 7 - 8	Manuscript drafting and submission to IEEE Conference	Submitted paper (IEEE TALE / EDUCON)
Next Semester	Phase 2 replication (CFA, Hou et al. validation, full journal paper)	Scopus Q1 Journal (IEEE TLT / C&E: AI)

6. Scope & Advisor Collaboration

Scope Boundary

The ORI metric is designed specifically for programming assignments with deterministic unit testing (Computer Science). Generalization to non-computing fields is explicitly noted as future work.

What We Need from Advisor

1. Approval to run S-SPARC in 1 active lab class (35-50 students).
2. Ethics review support & OSF co-sponsorship.
3. Joint mentorship and co-authorship on the resulting papers.

Key References (IEEE Style)

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